

PROBLEM 1: MOVING AVERAGES

Consider $x_0, x_1, \dots$ and two running averages defined as, for $0 < \beta < 1$ and $\gamma > 0$.

$$\begin{aligned} y_0 &= 0 \\ y_{t+1} &= \beta y_t + (1 - \beta)x_t \end{aligned}$$

$$\begin{aligned} z_0 &= 0 \\ z_{t+1} &= \beta z_t + \gamma x_t \end{aligned}$$

Part (a). Suppose x_t i.i.d. from distribution with mean vector $\bar{x}$ Give values for

$$\bar{y} = \lim_{t \rightarrow \infty} \mathbb{E} y_t$$

and

$$\bar{z} = \lim_{t \rightarrow \infty} \mathbb{E} z_t$$

as functions of β, γ , and $\bar{x}$.

Proof.

$$\begin{aligned} \mathbb{E} y_2 &= \beta \mathbb{E} y_1 + (1 - \beta)\bar{x} \\ &= \beta(\beta \mathbb{E} y_0 + (1 - \beta)\bar{x}) + (1 - \beta)\bar{x} \\ &= \beta^2 \mathbb{E} y_0 + \beta(1 - \beta)\bar{x} + (1 - \beta)\bar{x} \\ &= (1 - \beta)\bar{x} \sum_{i=0}^1 \beta^i \\ \implies \mathbb{E} y_t &= (1 - \beta)\bar{x} \sum_{i=0}^{t-1} \beta^i \\ \implies \bar{y} &= \lim_{t \rightarrow \infty} \mathbb{E} y_t = (1 - \beta)\bar{x} \sum_{i=0}^{\infty} \beta^i \\ &= \bar{x} \end{aligned}$$

For z_t , we have

$$\begin{aligned} \mathbb{E} z_3 &= \beta \mathbb{E} z_2 + \gamma \bar{x} \\ &= \beta(\beta \mathbb{E} z_1 + \gamma \bar{x}) + \gamma \bar{x} \\ &= \beta^2 \mathbb{E} z_1 + \beta \gamma \bar{x} + \gamma \bar{x} \\ &= \beta^2(\beta \mathbb{E} z_0 + \gamma \bar{x}) + \beta \gamma \bar{x} + \gamma \bar{x} \\ &= \beta^3 \mathbb{E} z_0 + \gamma \bar{x} \sum_{i=0}^2 \beta^i \\ \implies \mathbb{E} z_t &= \gamma \bar{x} \sum_{i=0}^{t-1} \beta^i \\ \implies \bar{z} &= \lim_{t \rightarrow \infty} \mathbb{E} z_t = \gamma \bar{x} \sum_{i=0}^{\infty} \beta^i \\ &= \frac{\gamma}{1 - \beta} \bar{x} \end{aligned}$$

□

Part (b). Express z_t as a function of y_t, β , and γ .

Proof. From part (a) we have:

$$z_t = \gamma \sum_{i=0}^{t-1} \beta^i x_{t-1-i}$$

$$y_t = (1 - \beta) \sum_{i=0}^{t-1} \beta^i x_{t-1-i}$$

This implies that

$$z_t = \frac{\gamma}{1 - \beta} y_t$$

□

PROBLEM 2: VARIANCE OF AN EXPONENTIAL MOVING AVERAGE

Consider a running average for computing $\hat{\mu}_1, \dots, \hat{\mu}_t$ from $x_1, \dots, x_t$

$$\hat{\mu}_0 = 0$$

$$\hat{\mu}_t = \left(1 - \frac{1}{N}\right) \hat{\mu}_{t-1} + \frac{1}{N} x_t$$

Part (a). Suppose x_t are i.i.d. with variance σ_x^2 . Find

$$\sigma_{\hat{\mu}, \infty}^2 = \lim_{t \rightarrow \infty} \text{Var } \hat{\mu}_t$$

Proof. Since $x_t \perp \hat{\mu}_{t-1}$, we have

$$\begin{aligned} \text{Var } \hat{\mu}_t &= \left(1 - \frac{1}{N}\right)^2 \text{Var } (\hat{\mu}_{t-1}) + \frac{1}{N^2} \text{Var } x_t \\ &= \left(1 - \frac{1}{N}\right)^2 \text{Var } (\hat{\mu}_{t-1}) + \frac{\sigma_x^2}{N^2} \end{aligned}$$

Taking limits on both sides, we have

$$\begin{aligned} \sigma_{\hat{\mu}, \infty}^2 &= \left(1 - \frac{1}{N}\right)^2 \sigma_{\hat{\mu}, \infty}^2 + \frac{\sigma_x^2}{N^2} \\ \left(1 - \left(1 - \frac{1}{N}\right)^2\right) \sigma_{\hat{\mu}, \infty}^2 &= \frac{\sigma_x^2}{N^2} \\ \left(1 - 1 + \frac{1}{N}\right) \left(1 + 1 - \frac{1}{N}\right) \sigma_{\hat{\mu}, \infty}^2 &= \frac{\sigma_x^2}{N^2} \\ \frac{2N - 1}{N^2} \sigma_{\hat{\mu}, \infty}^2 &= \frac{\sigma_x^2}{N^2} \\ \sigma_{\hat{\mu}, \infty}^2 &= \frac{\sigma_x^2}{2N - 1} \end{aligned}$$

□

Part (b). Compare $\sigma_{\hat{\mu}, \infty}^2$ to the variance of

$$\hat{\mu} = \frac{1}{N} \sum_{t=1}^N x_t$$

Proof.

$$\begin{aligned} \text{Var } \hat{\mu} &= \frac{1}{N^2} \sum_{t=1}^N \text{Var } x_t \\ &= \frac{\sigma_x^2}{N} \end{aligned}$$

□

PROBLEM 3. REFORMULATING MOMENTUM AS AN EXPONENTIAL MOVING AVERAGE

Consider the following update equation.

$$y_0 = 0$$

$$y_t = \left(1 - \frac{1}{N}\right) y_{t-1} + x_t$$

Part (a). Assume y_t converges to a limit. If the input sequence is constant $x_t = c$ what is the limit?

Proof. We have the following

$$y = \left(1 - \frac{1}{N}\right) y + c$$

$$y = Nc$$

□

Part (b). y_t is an exponential moving average of what quantity?

Proof. y_t is an exponential moving average of Nx_t

□

Part (c). Express y_t as a function of μ_t where μ_t is the exponential moving average of x_t with parameter $\frac{1}{N}$.

Proof. Recall that we have:

$$y_t = N \left(1 - \frac{1}{N}\right) \sum_{i=0}^{t-1} \frac{1}{N^i} x_{t-1-i}$$

On the other hand, we have

$$\mu_t = \left(1 - \frac{1}{N}\right) \sum_{i=0}^{t-1} \frac{1}{N^i} x_{t-1-i}$$

This gives us that

$$y_t = N\mu_t$$

□

PROBLEM 4. BIAS CORRECTION

Consider the following update equation for computing $y_1, \dots, y_t$ from $x_1, \dots, x_t$.

$$y_t = \left(1 - \frac{1}{t \wedge N}\right) y_{t-1} + \frac{1}{t \wedge N} x_t$$

If $x_t = c$ give a closed form solution for y_t .

Proof. Let $t < N$. We have:

$$\begin{aligned} y_t &= \left(1 - \frac{1}{t}\right) y_{t-1} + \frac{1}{t} c \\ &= \left(1 - \frac{1}{t}\right) \left(\left(1 - \frac{1}{t-1}\right) y_{t-2} + \frac{1}{t-1} c \right) + \frac{1}{t} c \\ &= \left(1 - \frac{1}{t}\right) \left(1 - \frac{1}{t-1}\right) y_{t-2} + \left(1 - \frac{1}{t}\right) \frac{1}{t-1} c + \frac{1}{t} c \\ &= c \prod_{i=2}^t \left(1 - \frac{1}{i}\right) + c \sum_{i=1}^t \frac{1}{i} \prod_{j=i+1}^t \left(1 - \frac{1}{j}\right) \end{aligned}$$

Alternatively,

$$y_1 = c$$

and for $t > 1$, we have that y_{t+1} is a convex combination of y_t and c . Since $y_1 = c$, we have that $y_t = c$ for all t .

□

PROBLEM 5.

Part (a). Consider vanilla SGD on cross entropy loss for classification with batch size 1 and no momentum. In this case, we have:

$$\Phi_{t+1} = \Phi_t - \eta \nabla_{\Phi} \log \Pr(y|x)$$

Suppose someone uses log base 2 and uses the update

$$\Phi_{t+1} = \Phi_t - \eta' \nabla_{\Phi} \log_2 \Pr(y|x)$$

Suppose we find that η works for the natural log update. What value of η' should be used in the second version?

Proof.

$$\Phi_{t+1} = \Phi_t - \eta' \nabla_{\Phi} \log_2 \Pr(y|x)$$

$$\Phi_{t+1} = \Phi_t - \frac{\eta'}{\log 2} \nabla_{\Phi} \log \Pr(y|x)$$

So we let $\eta' = \eta \log 2$ to get the same update as the natural log version. □

Part (b). Consider we have the following simplified version of RMSprop

$$\Phi_{-} = \frac{\eta}{\sigma_i} \nabla_{\Phi} \mathcal{L}_{\Phi}(x, y)$$

Where σ_i is the exact standard deviation of the i th coordinate of $\nabla_{\Phi} [i] \mathcal{L}_{\Phi}(x, y)$

If we replace $\mathcal{L}$ by $2\mathcal{L}$ what learning rate η' should we use to get the same temperature?

Proof. We compute σ_i as follows:

$$\sigma_i'^2 = \mathbb{V}\text{ar}(\nabla_{\Phi} [i] 2\mathcal{L}_{\Phi}(x, y)) = 4 \mathbb{V}\text{ar}(\nabla_{\Phi} [i] \mathcal{L}_{\Phi}(x, y)) = 4\sigma_i^2$$

So, $\sigma_i' = 2\sigma_i$. This gives us the following update:

$$\begin{aligned} \Phi_{-} &= \Phi_t[i] - \frac{\eta'}{\sigma_i'} \nabla_{\Phi} [i] 2\mathcal{L}_{\Phi}(x, y) \\ &= \Phi_t[i] - \frac{\eta'}{2\sigma_i} 2\nabla_{\Phi} [i] \mathcal{L}_{\Phi}(x, y) \\ &= \Phi_t[i] - \frac{\eta'}{\sigma_i} \nabla_{\Phi} [i] \mathcal{L}_{\Phi}(x, y) \end{aligned}$$

Therefore, we just take $\eta' = \eta$ to get the same update as before. □

ADAPTIVE SGD

Consider a generalization of RMSprop where we allow an arbitrary adaptation with different learning rates for different parameter values Specifically:

$$\Phi_{t+1} = \Phi_t - \eta(A(\Phi_t, x_t, y_t) \odot \nabla_{\Phi} \mathcal{L}(\Phi_t, x_t, y_t))$$

Consider the special case

$$\begin{aligned} A(\Phi, x, y) &= \frac{1}{\sqrt{s(\Phi, x, y) + \epsilon}} \\ s(\Phi, x, y) &= \frac{1}{d} \|\nabla_{\Phi} \mathcal{L}(\Phi, x, y)\|^2 \end{aligned}$$

where d is the dimension of Φ .

Part (a). Let Φ^* be a stationary point of the update equation, i.e. the expected update over a random draw from the population is zero. Write this stationarity condition on Φ^* explicitly as an expectation equaling zero

Proof.

$$\mathbb{E}_{(x,y) \sim \text{Pop}} \frac{\nabla_{\Phi} \mathcal{L}(\Phi^*, x, y) \sqrt{d}}{\|\nabla_{\Phi} \mathcal{L}(\Phi^*, x, y)\| + \epsilon \sqrt{d}} = 0$$

□

Part (b). Is Φ^* a stationary point of the original loss - a point where the expected gradient of $\mathcal{L}(\Phi^*, x, y)$ is zero?

Proof. No. A weighted average equalling zero does not imply the unweighted average is zero. $\square$

Part (c). Do these observations have implications for the adaptive methods described?

Proof. Yes. This implies that adaptive methods are not guaranteed to converge to stationary points of the original loss. $\square$

PROBLEM 7.

Consider the following multiplicative update SGD, for batch size 1

$$\Phi_{t+1}[i] = \Phi_t[i] - \eta(\epsilon \vee |\Phi_t[i]|)\hat{g}(\Phi, x_t, y_t)[i]$$

For $|\Phi_t[i]| \geq \epsilon$, this is a multiplicative update.

Part (a). At a stationary point of the loss function, is the expected update over a random draw of (x, y) always equal to zero?

Proof. Let Φ^* be a stationary point of the loss function. Then, we have

$$\mathbb{E}[\hat{g}] = \mathbb{E}[\nabla_{\Phi} \mathcal{L}(\Phi^*, x, y)] = 0$$

Yes. $\square$

Part (b). Consider an adaptive algorithm which makes the update proportional to the loss.

$$\Phi_{t+1} = \Phi^t - \eta \mathcal{L} \hat{g}_t$$

Is a stationary point of the loss function always a stationary point of the update?

Proof. Again, an unweighted average equalling zero does not imply the weighted average is zero. $\square$